S.No: 2	Exp. Name: *Write a Program to Search an element using Binary Search and Recursion*	Date:2025-04-04

Aim:

Write a program to `search` the given element from a list of elements with `binary search` technique using **recursion**.

Note: Write the functions **read()**, **bubbleSort()**, **display()** and **binarySearch()** in `Program912a.c`

Source Code:

Program912.c

```
#include <stdio.h>
#include "Program912a.c"
void main() {
   int a[20], n, key, flag;
   printf("Enter value of n : ");
   scanf("%d", &n);
   read1(a, n);
   bubbleSort(a, n);
   printf("After sorting the elements are : ");
   display(a, n);
   printf("Enter key element : ");
   scanf("%d", &key);
   flag = binarySearch(a, 0, n - 1, key);
   if (flag == -1) {
      printf("The given key element %d is not found\n", key);
   } else {
      printf("The given key element %d is found at position : %d\n", key, fla
g);
   }
}
```

Program912a.c

```
#include <stdio.h>

// Function to read n elements into array a
void read1(int a[], int n) {
    int i;
    printf("Enter %d elements : ", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
}

// Function to sort the array using Bubble Sort
void bubbleSort(int a[], int n) {
    int i, j, temp;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(a[j] > a[j + 1]) {
```


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```
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
}

// Function to display the array elements
void display(int a[], int n) {
    int i;
    for(i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    printf("\n");
}

// Recursive function to perform Binary Search
int binarySearch(int a[], int low, int high, int key) {
    if(low > high) {
        return -1;  // key not found
    }
    int mid = low + (high - low) / 2;
    if(a[mid] == key) {
        return mid;
    }
    else if(key < a[mid]) {
        return binarySearch(a, low, mid - 1, key);
    }
    else {
        return binarySearch(a, mid + 1, high, key);
    }
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter value of n : 5
Enter 5 elements : 33 55 22 44 11
After sorting the elements are : 11 22 33 44 55 11
Enter key element : 11
The given key element 11 is found at position : 0

Test Case - 2
User Output
Enter value of n : 4
Enter 4 elements : 23 67 45 18
After sorting the elements are : 18 23 45 67 24
Enter key element : 24
The given key element 24 is not found